



Effects of Joint Video Modeling on Unscripted Play Behavior of Children with Autism Spectrum Disorder

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Abstract

Preschool aged children with autism spectrum disorder (ASD) have marked deficits in pretend play that impede interactions with typically developing peers in inclusive early childhood settings. This study aimed to teach three young children with ASD to engage in pretend play behaviors with their peers. A multiple probe across participants experimental design was used to evaluate the effects of joint video modeling on scripted and unscripted verbalizations and scripted and unscripted play actions of children with ASD. The participants showed improvement on unscripted verbalizations during pretend play with typically developing peers in an inclusive early childhood setting.

Keywords Peer-mediated intervention · Pretend play · Video-modeling

Early childhood curricula emphasize play as an activity to promote social, emotional, cognitive, and physical development (Milteer and Ginsburg 2012). Most early childhood classrooms are structured to provide ample opportunities for play, particularly pretend play, because it allows social interactions that are essential to children's development (Pellegri 2009). Pretend play is defined as using an object to represent a different object, performing actions with objects in a pretend manner (e.g., feeding a doll), or giving animacy to objects (e.g., making a doll go down a slide; Lifter et al. 2005). Pretend play can often involve social components such as orienting towards others or what they are doing, physical contact, role-taking, and turn-taking (Power 2000; Yang et al. 2003). Pretend play can also involve the use of language such as narrating actions, commenting on the actions of others and giving animacy to objects (Lifter et al. 2005).

Children with autism spectrum disorder (ASD) demonstrate delays in functional and spontaneous pretend play compared to same aged peers, which might contribute to delays in social development (Jarrold 2003). When children with ASD do engage in pretend play, it occurs at lower rates

and involves less language compared to typically developing peers (Amato et al. 1999; Charman and Baron-Cohen 1997; Jarrold 2003; Lewis and Boucher 1995; Wing et al. 1977). Teaching pretend play behaviors is important as it increases the likelihood of peer interactions (McConnell 2002). In addition, the ability to engage in pretend play has been associated with gains in social-communication in children with ASD (Amato et al. 1999; Hobson et al. 2012) and therefore has been recommended as a direct focus of early intervention programs (Kasari et al. 2006).

Inclusive early childhood settings, in part given their emphasis on play for all children, are said to offer opportunities for children with ASD to interact with typically developing peers (Gena 2006). Early childhood settings provide children multiple opportunities for play that are child-led, open-ended, and varied. Because children with ASD generally do not enter early childhood programs with the same play skills as typically developing children, they may not be as prone to demonstrate gains in social, emotional, cognitive, and physical domains simply by accessing play opportunities (Gutierrez et al. 2007). Children with ASD require structured supports and explicit interventions to acquire appropriate play skills in order to more fully participate in and benefit from early childhood educational environments (Barton and Pavilanis 2012; DiCarlo and Reid 2004; Kasari et al. 2006). Few early childhood settings provide such explicit instruction (Hess et al. 2008).

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One intervention that early childhood special educators commonly report using and being knowledgeable about is peer mediated intervention (PMI; Paynter et al. 2017). In PMI, peers are carefully selected and systematically trained to serve as intervention agents who can promote social interaction with children with ASD (Odom et al. 1999; Odom and Strain 1986). PMI is an effective strategy for improving the social-communication of preschool children with ASD in inclusive settings during play (Barber et al. 2015; Laushey and Heflin 2000; also see; Watkins et al. 2015 for a review) and an ideal strategy for promoting generalization and maintenance (Goldstein et al. 2007; McFadden et al. 2014). Though PMI has shown to improve the social-communicative responses of children with ASD directed towards peers, it has not yet been evaluated for social-communication during pretend play interactions (e.g., narrating objects and giving animacy to objects). Teaching children with ASD to engage in pretend play interactions with typically developing peers in early childhood environments involves more advanced language and social development (e.g., social cognition, emotional regulation, and affect; Guralnick 1992) that requires explicit and systematic instruction (e.g., video modeling, visual or written scripts, self-management, social stories, and pivotal response training; see Barton and Wolery 2008; Camargo et al. 2016). Although teaching peers to interact with children with ASD, as in PMI, may be beneficial for peers and may promote maintenance and generalization, children with ASD also need explicit training (e.g., prompting, reinforcement) in order to play with their peers (Goldstein et al. 2007; McFadden et al. 2014).

A widely used intervention to teach scripted verbalization during pretend play to children with ASD is video modeling (D'Ateno et al. 2003; MacDonald et al. 2005; MacManus et al. 2015; Palechka and MacDonald 2010). Video modeling has involved typically developing peers, referred to here as *joint video modeling* (JVM), to teach children with ASD and typically developing peers to play together with various materials. Though limited, these studies have demonstrated positive outcomes with JVM for teaching play skills to children with ASD and their typically developing peers or siblings (MacDonald et al. 2009; Maione and Mirenda 2006; Reagon et al. 2006).

Reagon et al. (2006) were one of the first to conduct an evaluation of JVM to teach scripted statements and play actions to a 4-year-old boy with ASD and his sibling in their home. The results suggested JVM may have been effective for teaching a variety of scripted responses across four different play-sets that maintained during follow-up sessions. Although methodological limitations prevent a causal analysis, the results suggested that JVM had promise for teaching scripted play to children with ASD.

MacDonald et al. (2009) experimentally evaluated JVM on scripted verbalizations and play actions of two preschool

aged children with ASD, and two typically developing peers. The researchers created videos that depicted adults modeling functional play actions with verbalizations, and presented the video models to the children with ASD and an assigned typical peer partner. In a multiple probe across play-sets design, the researchers demonstrated a functional relation between JVM and scripted play actions and verbalizations, without the additional use of prompts and reinforcement.

Although JVM has been effective for teaching scripted play responses, children with ASD are at risk for engaging in rote responding and require explicit instructional protocols to demonstrate more natural play interactions across varied contexts (Kasari et al. 2006). Researchers who have been successful at teaching scripted pretend play have suggested future video modeling research might incorporate multiple video exemplars to improve the unscripted responses of participants (D'Ateno et al. 2003). Using multiple exemplars within video modeling might involve showing a child several videos of the same general behaviors (e.g., playing with toys) while strategically varying aspects of the model. For example, researchers might vary the toys in the video model from one instructional session to another so that the child learns to play with a range of toys as opposed to a single toy. Alternatively, researchers might vary comments a child in a video makes while playing with toys to teach the learner several potential response options when presented with similar toys (see Plavnick and Dueñas 2018). Programming multiple exemplars is an instructional design strategy that provides a range of examples so that learners acquire repertoires as opposed to rote responses—it is an essential component of effective instruction designed to promote generalization (Stokes and Baer 1977).

Recent studies examining video modeling have used multiple video exemplars to attempt to teach unscripted verbalizations to children with ASD. MacManus et al. (2015) created video models that arranged three play scenarios with three play-sets in varied combinations in order to create flexible responding among participants. After training, all participants learned verbalizations and actions depicted in videos, and combined verbalizations and actions across videos to demonstrate a specific type of response generalization called recombinative generalization (Goldstein and Moussetis 1989). These results demonstrate the effectiveness of multiple exemplar video modeling for *solitary* pretend play and offer an intervention with potential to teach children with ASD to engage in pretend play with peers.

To our knowledge, only one study has evaluated the effects of multiple exemplar video modeling on unscripted responses of children with ASD with typical peers. Maione and Mirenda (2006) taught a 5-year-old boy with ASD to engage in unscripted play actions and verbalizations using multiple exemplar video modeling with a peer partner. The researchers created three videos for each of the

three play-sets and conducted video modeling sessions daily. However, a functional relation was not established between JVM and play actions/verbalizations because the child engaged in a high incidence of repetitive behaviors with one of the play-sets. JVM alone, therefore, was not determined successful for this participant. Though multiple exemplar videos have potential for promoting spontaneous pretend play for children with ASD, additional research is needed to explore the effects of multiple exemplar videos in an inclusive preschool setting and with additional children with ASD.

Therefore, the purpose of the current study was to extend the research on JVM to teach pretend play to children with ASD in a natural setting by programming multiple exemplars to specifically evaluate unscripted responses. During baseline, we assessed the effects of typical peers' prior exposure to PMI and video modeling to evaluate its effect on social-communication for children with ASD.

Method

Participants with ASD

Participants with ASD included Rita, Saul, and Lucas who were 4 years old and attended an early intensive behavior intervention program housed within a child development laboratory preschool at a Midwestern University. The participants were admitted to this program based on an outside diagnosis of ASD made by an independent psychologist using the *Autism Diagnostic Observation Schedule* (Lord et al. 2000). Participants spent approximately 5 h per day in an intensive applied behavior analysis therapy room and 2.5 h in an inclusive early childhood classroom with typically developing 4-year-old children.

Children with ASD were included in the present study if they participated in the early childhood classroom during free-choice, spoke using a minimum of three-word phrases, and showed low levels of social interactions with peers as determined by the *Verbal Behavior Milestones Assessment and Placement Program* (VB-MAPP; Sundberg 2008). The VB-MAPP was administered 1 month prior to the study by the first author.

According to the VB-MAPP, Rita imitated the gross motor actions of her peers during large and small group activities with prompting. She interacted with peers during adult-led activities but did not interact with peers during free play or any other child-led activities (e.g., recess). Saul was able to spontaneously engage in parallel play with peers during preferred activities (e.g., blocks and sand box). He also spontaneously imitated the gross motor movements of peers during large group activities. According to the VB-MAPP, Lucas spontaneously engaged in parallel play with peers

during preferred activities (e.g., blocks and sand box). He also spontaneously imitated the motor movements of peers. However, Lucas seldom made eye contact with peers.

Typically Developing Peers

Typically developing peers included three 4-year old children; two girls (Emma and Jaya) and one boy (Mason) who attended the same child development laboratory as the participants with ASD. The first author asked the director of the preschool program and head teacher to select peers that fulfilled the following prerequisites based on previous PMI research (see Katz and Girolametto 2013): (a) the child had a history of either positive or neutral interactions with children with ASD, (b) the child demonstrated age appropriate social skills, (c) the child followed adult directives, (d) the child was well liked by other peers, and (e) the child demonstrated consistent school attendance. Emma was paired with Rita, Jaya was paired with Saul, and Mason was paired with Lucas.

The first author trained all peers at the beginning of the school year to be "play partners" for children with ASD and to initiate play using video modeling. The training was adapted from Katz and Girolametto (2013) by training the typical peer and child with ASD in the classroom rather than outside of the classroom and by using video modeling to train typical peers. This was done to accommodate a request made by the lead teacher to reduce the amount of time children were removed from the classroom. The peer training included the following: (a) introducing the children with ASD using photographs of the children, the children's names, and their two favorite activities and toys, (b) making a brief statement about children with ASD's challenges playing with others, and (c) reading a children's book about accepting differences in others. Then peers were taught to make simple initiations to play using video modeling and role-play. Finally, the researcher administered joint play training to typical peers and children with ASD in the inclusion classroom. Three 15-min sessions were administered each day and involved early childhood educators mediating interactions via prompting over a 12-week period.

Setting

All assessment and intervention sessions were conducted in the inclusive early childhood classroom. Two behavior technicians, assigned to deliver applied behavior analytic therapy to children with ASD, were present in the inclusive classroom. One general education teacher and two student teachers were assigned to 15 typically developing children in the classroom. The room was organized in stations that included a section for large group, four tables for small group instruction, a reading area, a pretend play area, and a snack table. All sessions were conducted with an experimenter present.

Measurement

All sessions were video-recorded and frequencies of the following behaviors were collected from videos for all participants: (a) scripted verbalizations, (b) scripted play actions, (c) unscripted verbalizations of two or more words, and (d) unscripted play actions. The measurement system was adopted and expanded from MacDonald et al. (2009).

The primary dependent variables of interest were the frequency of unscripted verbalizations and play actions. Unscripted verbalizations were defined as any vocalization that was not modeled in the video but that expanded on the current verbalizations related to play. These were made up of the following: (a) a response to a previous comment or question made by the peer partner, (b) vocally labeling actions with toys, (c) vocally labeling actions performed by the peer model, and (d) vocally labeling an object within the play-set. An example of labeling own actions was using the staircase on the tree house and saying, "Going up the stairs." Alternatively, an example of labeling actions performed by the peer model was seeing the peer swing a toy and saying, "He is swinging." These were only scored once for a particular interaction, unless they were produced under new or different circumstances. For example, saying "Oh no, fell down!" when the toy bunny fell off the slide; and then later saying, "Oh no, fell down!" when the toy bunny fell off the hammock. Repetitive verbalizations unrelated to the context were not scored.

Unscripted play actions were defined as any play action not seen in the video that expanded on the current play and were made up of the same components described for scripted play including, functional play actions involving the character as agent, and functional play actions directed to the toy character. Unscripted play was scored if an action did not have one to one correspondence with the video model or if they were from previous videos not shown during that intervention session. Unscripted play actions were only counted once and not counted if the action was repetitive or involved motor stereotypies, such as making the bunny jump repetitively.

The second dependent variable of interest was the percentage of correct imitations to the video model (i.e., scripted verbalizations and play actions). Scripted verbalizations were defined as any verbalizations that matched the verbalizations within the video modeling script. Although such verbalizations were unlikely during baseline, we isolated this as a dependent measure to assess the direct impact of presenting a scripted interaction via video model during intervention. Scripted verbalizations were scored if the child vocally imitated the same phrase as the projected model within the video modeling script. If the child omitted a word or made a slight change in a word, for example, saying "going sleep" versus "going to sleep," the response

was scored as a correct scripted verbalization as long as two out of three words were emitted. Scripted verbalizations were not counted more than once or if they occurred out of sequence.

Scripted play actions were defined as any play action imitated by the child from the video model. These were made up of functional play actions involving the character as agent (e.g., pretending the toy bunny spilled his pizza) and functional play actions directed to the toy character (e.g., feeding the toy bunny). They were counted even if they were non-identical but served the same function. For example, placing the bunny down the slide and letting it go versus holding the bunny all the way down the slide.

Interobserver Agreement

The first author trained an undergraduate research assistant who was blind to the purpose of the study to code dependent variables from training videos. After reaching 90% reliability with the first author on training videos, the research assistant coded 100% of baseline and intervention videos while the first author coded 40% of randomly selected videos evenly distributed across all conditions and participants to assess interobserver agreement (IOA). Both observers recorded dependent measures and the time in the video when events occurred. An event was scored as an agreement if both observers recorded an instance of the same dependent measure within 3 s of one another. An event was scored as a disagreement if the observers recorded the time the behavior occurred or the type of behavior differently. Percentage of agreement was calculated for each dependent variable by dividing the number of total agreements by the number of total possible agreements plus disagreements and multiplying by 100. Mean IOA between the first author and the research assistant during baseline was 98.83% (range 93–100%) and 100% for unscripted and scripted verbalizations, and 93% (range 84–100%) and 98.33% (range 90–100%) for unscripted and scripted play actions. Mean IOA between the first author and the research assistant during intervention was 86% (range 81–95%) and 91% (range 85–100%) for unscripted and scripted verbalizations, and 88.17% (range 78–100%) and 91.33% (range 80–100%) for unscripted and scripted play actions.

Experimental Design

A single-case multiple-probe across participants design (Horner and Baer 1978) was used to assess the effects of JVM on acquisition of scripted and unscripted verbalizations and play actions by children with ASD. In a multiple probe across participants design, visual analysis is used to determine a functional relation through systematic and sequential manipulation of the independent variable. That

is, the independent variable is sequentially introduced across participants, while continuing to collect baseline data for remaining participants to ensure change in the dependent variable occurs only when introducing the independent variable. Data are visually analyzed for stability of the dependent variable during baseline and change in the dependent variable is assessed based on magnitude of effect. The effect is determined by visual inspection of the immediacy of the effect, the pattern of that change (i.e., trend), and the overall level of change (e.g., mean difference) as compared to baseline.

The multiple probe design across participants is similar to a multiple baseline across participants, except that the researcher administers intermittent probes during baseline rather than continuous measurement of the dependent variable (Gast and Ledford 2014). In this study, using a multiple-probe design minimized the amount of time children were asked to play together during baseline without intervention for the children with ASD, while allowing for a demonstration of a functional relation between JVM and the dependent measures.

Procedures

Materials

At the start of the study teachers and behavior technicians were asked to rank order eight play-sets based on what they believed were most preferred for each dyad (i.e., pairing of typical peer and child with ASD). A *Calico Critters*® play-set was rated highest and used during the imaginative play sequences for all dyads. The play-set was the Lakeside Lodge, which included a hot tub, two hammocks, a ladder, a table and two chairs, a stove, and a refrigerator. Researchers used a *Hero 4 Go Pro*™ action camera to film three different play scripts involving the play-set. Videos were later loaded onto an *Apple iPad* in an application called *My Pictures Talk* for viewing. Children used headphones to listen to the videos, with a splitter to allow both members of a dyad to listen to the video at the same time.

Three video clips of adults playing with the play-set were used during the video modeling phases of the study. Two adult models were recorded acting out the predetermined play sequences (scripts and sequences are available from the first author upon request). The models in the video remained in the same position in order to help the children attend to the model they were assigned (see MacDonald et al. 2009). Female and male models were selected to create the 30 s videos. The three video exemplars showed adults' hands acting as agents for the toy bunnies. The videos varied with regards to the modeled play actions, modeled verbalizations, and stimuli (e.g., mama bunny and sister bunny vs. brother and sister bunny). The play sequences were derived from

observations of typically developing children in the early childhood classroom playing with the same play-set.

Baseline

The purpose of the baseline probe was to assess the frequency of verbalizations and play actions after typically developing peers had received PMI training for 12 weeks and assess the effects of typical peer exposure to video modeling on responses of children with ASD. The number of baseline probes varied across participants due to the design requirements of a multiple probe across participants design (see “[Experimental Design](#)” section). During baseline probes, the video model was presented to the typical peer only. This condition was used to evaluate the effects of the video modeling intervention for typically developing peers alone on the verbalizations and play actions of both participants in each dyad.

The typical peer was approached by the first author and asked to “Come watch a video.” The typical peer viewed the video on an iPad with headphones. The first author showed the video twice and redirected the peer to watch the video if he or she looked away. After the typical peer viewed the video twice, they were instructed to invite the child with ASD to play. Once both children were at the table, the typical peer was instructed to “Do and say what you saw in the video,” and the children were given 3 min to play with the play-set that was already at the table. If the children attempted to leave the play area they were physically prompted to play, although this did not occur.

Researchers presented three video exemplars, one in each session, to the typical peer during baseline to increase the likelihood of novel responding (MacManus et al. 2015; Maione and Mirenda 2006). Video exemplars were rotated across sessions to ensure each exemplar was presented as close to an equal number of times as possible, given the unknown number of intervention sessions. For example, during the first baseline probe the child saw the first video, during the second baseline probe the child saw the second video, and during the third baseline probe the third video was shown. This sequence was repeated for the following sessions.

Intervention

Intervention was similar to baseline except that both the typical peer and child with ASD were instructed to jointly watch the 30 s video model; these were the same videos as in baseline. After the request to invite the child with ASD to play, the children were seated at a small table side by side according to the model they were imitating in the video. The iPad was presented on the table directly in the middle of both children, who were seated next to each other and they were

exposed to the audio in the video using headphones with a splitter. The video was shown twice, and the researcher ensured their bodies were positioned in front of the screen and their faces were directed to the screen. If children looked away from the screen, the first author gesturally and verbally prompted them to look at the screen by pointing and saying, “Watch the video”. This was done to increase the likelihood that they were attending to the video. As in baseline, the three video exemplars were rotated across sessions to ensure equal number of presentations for each video exemplar. The children were instructed to “Do and say what you saw in the video” and were given 3 min to play with the toys that were set up at the table. Although this did not occur, if the children attempted to leave the play area the first author planned to physically redirect them to the play area by placing her hands on their shoulders and gently guiding them back to the play area. No additional prompts were provided.

Generalization Probes

Generalization probes were conducted once prior to intervention and on average after every third or fourth intervention session. Generalization probes were similar to the conditions described in baseline except that neither of the children was exposed to the video models and the child with ASD interacted with a different play partner but one that had participated in the study. A typical peer from a different dyad was instructed to invite the child with ASD to play. Once the typical peer invited the child with ASD and brought them to a table that was set up with the play-set, both children were instructed to play. That is, the researcher said, “Play with your friend.” The children were given 3 min to play with the toys. No additional instruction was provided.

Procedural Integrity for Video Modeling

Researchers created and used a checklist that consisted of the steps for implementation of video modeling to assess the integrity of implementation for baseline probes and intervention phases. For baseline probes, the checklist consisted of the following steps initiated by the adult: (a) located application on the iPad called My Pictures Talk® and opened correct video, (b) ensured video and audio worked, (c) approached peer model in the classroom and showed the video to the peer model by saying, “Watch this”, (d) said, “Go find [child’s name] and do and say what you saw in the video”, (e) showed the video twice, (f) did not provide any additional vocal instructions, and (g) remained behind the children.

For intervention, the checklist consisted of the following steps initiated by the adult: (a) located application on the iPad called My Pictures Talk® and opened correct video, (b) ensured video and audio worked, (c) approached peer model

in the classroom and instructed them to invite child with ASD to play, (d) once children were at play area placed the iPad in front of both peers and seated them in the same position of the model in the video, (e) showed the video twice, (f) said “It’s time to play, do and say what you saw in the video”, (g) remained behind the children, and (h) provided no further prompting or reinforcement.

The first author randomly selected 33% of videos to code procedural integrity and trained the secondary observer to 90% coding reliability from videos that were not part of the randomly selected sample. The mean percentage of steps implemented accurately during baseline was 100%. The mean percentage of steps implemented accurately during intervention was 92% (range 87–100%).

Results

Children with Autism

Unscripted Verbalizations

Figure 1 depicts the unscripted verbalizations during baseline, intervention, and generalization probes for Rita, Saul, and Lucas. Rita did not engage in unscripted verbalizations during baseline sessions. Her total unscripted verbalizations increased after two intervention sessions, with an overall mean of 2 (range 0–6) during the intervention phase. Despite variability in overall amount, Rita reliably performed some unscripted verbalizations beginning with the third intervention session. She also demonstrated generalization of unscripted verbalizations to a peer not in her dyad during intervention.

Saul displayed very low levels of unscripted verbalizations ($M=0.6$, range 0–2) during baseline. During intervention, total unscripted verbalizations were variable, but increased to an overall mean of 6 (range 3–10). Unscripted verbalizations generalized to a peer not in his dyad during intervention but did not remain at the same level ($M=3.75$, range 2–6).

Lucas displayed low levels of unscripted verbalizations during baseline ($M=0.3$, range 0–2). During intervention, total unscripted verbalizations increased to a mean of 3 (range 0–6). Unscripted verbalizations generalized to a peer not in his dyad during intervention but were low ($M=1.75$, range 1–3).

Unscripted Play Actions

Figure 2 depicts unscripted play actions during baseline, intervention, and generalization probes for Rita, Saul, and Lucas. Rita demonstrated unscripted play actions during baseline ($M=3.5$, range 1–5). During intervention,

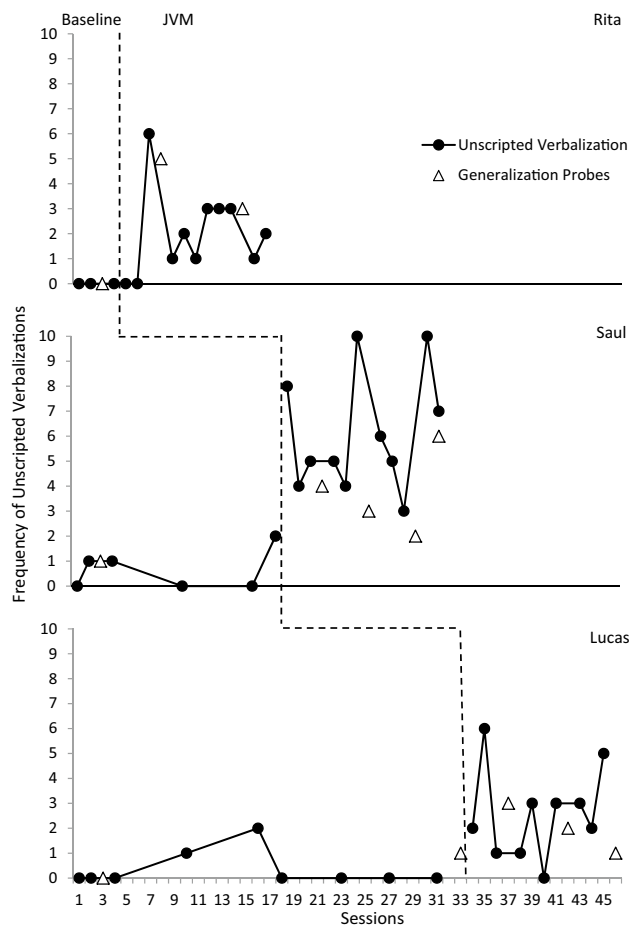


Fig. 1 Total unscripted verbalizations of children with ASD during baseline, intervention, and generalization

her unscripted play actions remained at the same level ($M = 3.6$, range 1–6). Generalization probes with a peer not in her dyad indicated the same levels of unscripted play actions.

Saul demonstrated low levels of unscripted play actions during baseline ($M = 2$, range 0–3). During intervention, his unscripted play actions increased to a mean of 7.7 (range 6–10). Generalization probes with a peer not in his dyad indicated slightly higher levels of unscripted play actions ($M = 8.75$, range 8–10).

Lucas demonstrated some unscripted play actions during baseline ($M = 3.3$, range 1–5), with very little play actions occurring during the final two baseline sessions. During intervention, his unscripted play actions initially increased from baseline levels, though were highly variable with a similar overall mean from baseline ($M = 3.75$, range 1–7). Generalization probes with a peer not in his dyad indicated similar patterns to intervention of unscripted play actions ($M = 3$, range 1–7).

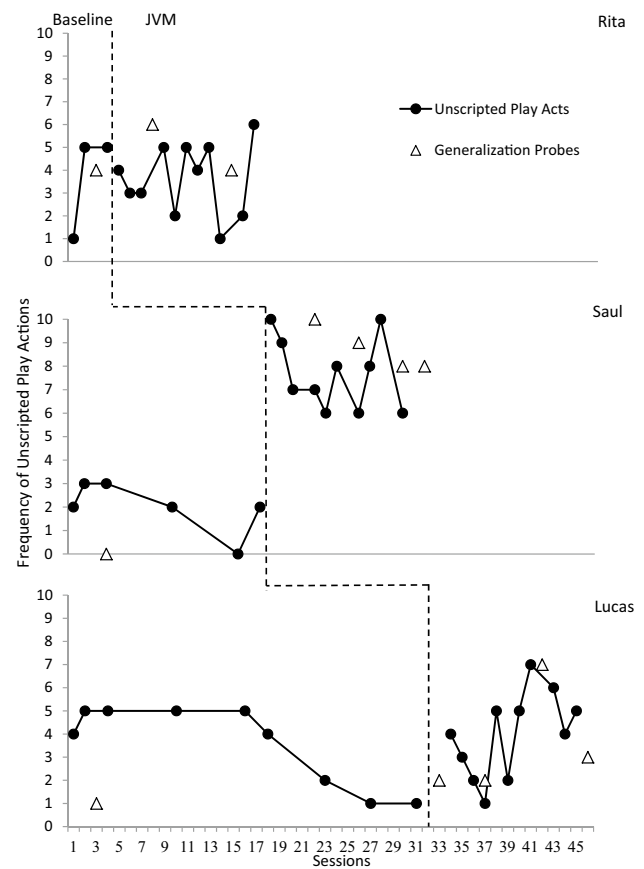


Fig. 2 Total unscripted play actions of children with ASD during baseline, intervention, and generalization

Scripted Verbalizations and Play Actions

Figure 3 depicts total percentage of scripted responses for Rita, Saul, and Lucas during baseline and intervention. The total possible scripted responses were eight and percentage of correct scripted responses is displayed on the y-axis in order to show overall adherence to the video models. Although a combined percentage does not allow a comparison of one type of responding to the other (e.g., high rates of play actions and low verbalizations), each participant engaged in scripted play actions at a percentage similar to verbalizations.

Rita performed one scripted play action during the second baseline session, with no scripted play actions during the first or third baseline sessions. During intervention, her total scripted verbalizations and play actions increased to a mean of 41% (range 29–63%). Rita demonstrated an increase in responding to the video model on the third session, where she achieved 63%. However, she did not achieve mastery criteria (i.e., 75% across two sessions).

Saul did not perform any scripted responses at baseline. Post intervention, his total scripted verbalizations and play

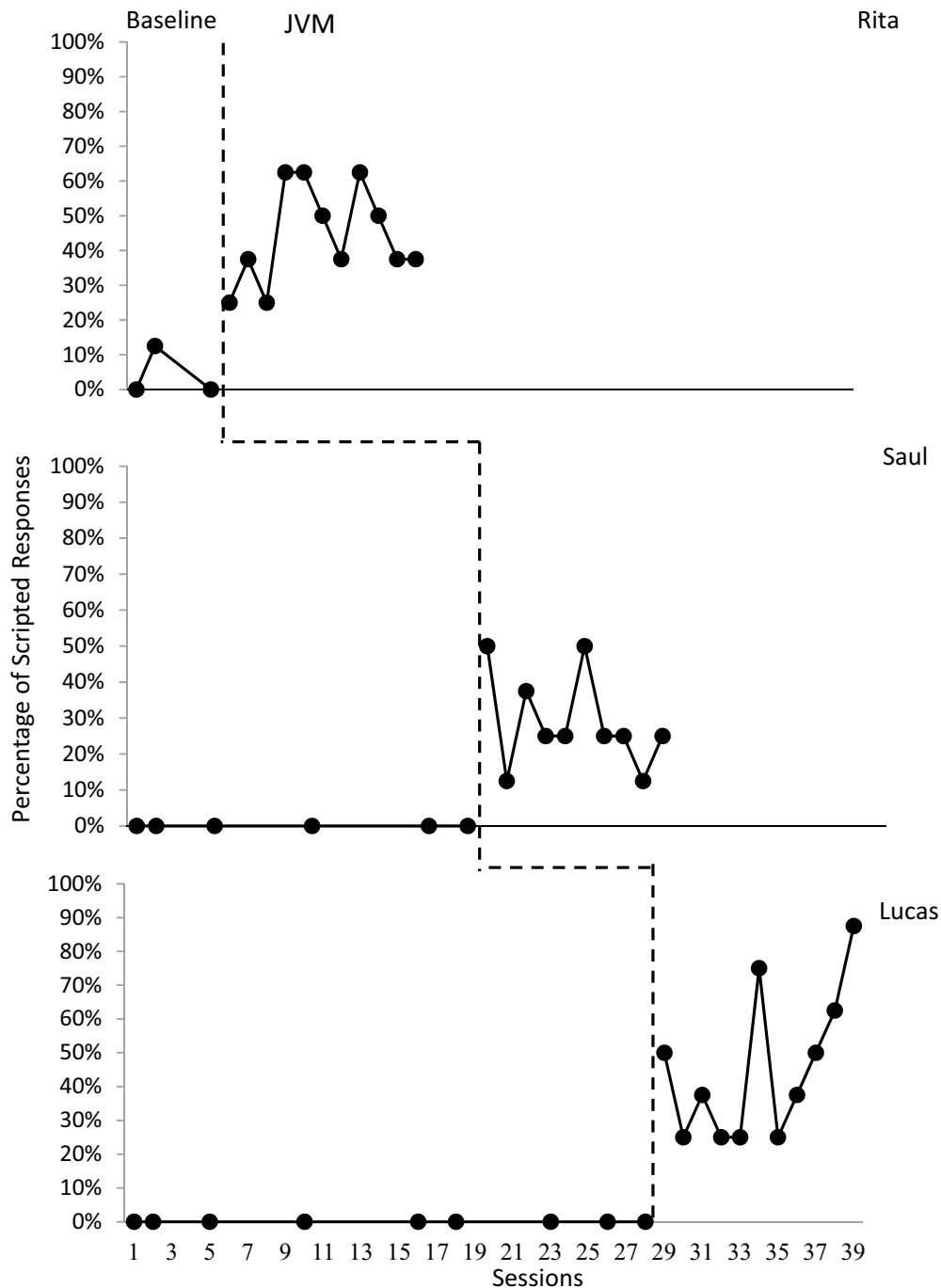


Fig. 3 Percentage of scripted responses of children with ASD during baseline, intervention, and generalization

actions increased to a mean of 28% (range 0–63%). Saul demonstrated an immediate response to the video model, imitating 63% of verbalizations and play actions on the first session, but responding gradually decreased over time. He did not achieve mastery criteria for scripted responses.

Lucas did not perform any scripted responses at baseline. Post intervention, his total scripted verbalizations and play actions increased to a mean of 44% (range 10–80%).

Lucas demonstrated an increase in responding to JVM on the 10th session, where he achieved 75%. Responding was somewhat variable, though generally followed an increasing trend. Lucas met mastery criteria for imitating the video model during the 11th session.

The average frequency of scripted verbalizations was low for all participants. Rita verbally scripted an average of 1.8 times versus 2 for unscripted. Saul verbally scripted

an average of 1 time versus 6 for unscripted. Lucas verbally scripted an average of 1 time versus 3 for unscripted.

Effect Size Calculations

To supplement visual analysis of data for participants with ASD, the authors calculated the effect of the intervention on scripted and unscripted verbalizations, and scripted and unscripted play actions with a web-based Tau-U calculator (see <http://www.singlecaseresearch.org/calculators/tau-u>). Tau-U offers non-parametric effect sizes by calculating the nonoverlap of data between baseline and intervention phases whilst controlling for possible trends during baseline (Parker and Vannest 2009; Rakap 2015). The weighted Tau-U for verbalizations were both medium-to-large (Rakap 2015) at 0.78 for scripted and 0.89 for unscripted with 95% CIs [0.44, 1] and [0.55, 1], respectively. The weighted Tau-U for play actions were medium-to-large (Rakap 2015) for scripted play actions at 0.82 but small (Rakap 2015) for unscripted play actions at 0.34 with 95% CIs [0.48, 1] and [0.02, 0.70], respectively.

Discussion

The purpose of this study was to examine the effects of a multiple exemplar JVM intervention on pretend play behavior (i.e., unscripted/scripted verbalizations and play actions) of children with ASD with their typically developing peers in an inclusive early childhood setting. Prior to JVM, participants with ASD engaged in low frequencies of verbalizations (range 0–2) with typically developing peers and demonstrated some appropriate pretend play actions with toys (e.g., feeding bunny, range 0–5). Upon introducing JVM, all children with ASD demonstrated increased levels of scripted and unscripted verbalizations during pretend play. Verbalizations persisted in the absence of video models. All participants also demonstrated an increase in scripted play actions with varied outcomes in unscripted play actions. The current investigation suggests JVM may be a viable intervention to teach unscripted verbalizations during pretend play to children with ASD, though more research is needed. The current study extends existing JVM research by demonstrating the emergence of unscripted verbalizations during pretend play within a natural environment with typically developing peers.

A functional relation between JVM and unscripted verbalizations by children with ASD was observed for all participants in this study, as unscripted verbalizations increased only when JVM was applied across all three participants. This outcome differed from some previous video modeling research wherein children acquired scripted but not unscripted responses (D'Ateno et al. 2003; MacDonald et al.

2009), though supported other research wherein a participant did demonstrate gains in unscripted responding (Maione and Mirenda 2006). Importantly, the participant in Maione and Mirenda (2006) demonstrated gains in unscripted verbalizations when video modeling was administered, though he often emitted 10 or more verbalizations during baseline sessions. Alternatively, participants in the present investigation engaged in near-zero levels of verbalization during baseline. The present investigation suggests multiple exemplar JVM can lead to emergence of unscripted verbalizations, even among children who demonstrate very little verbalization during baseline.

The pattern of unscripted play actions was markedly different than that of verbalizations with Saul demonstrating the only clear increase from baseline to intervention. Rita and Lucas demonstrated some play actions during baseline followed by a subsequent decrease. The observed gains in unscripted play actions when peers began watching the video model during baseline suggests that peer behavior (e.g., prompting and modeling) may produce some behavior change in children with ASD but may not be enough to produce clear and sustained change in play behaviors. Although Rita and Lucas demonstrated approximately five unscripted play actions across several baseline sessions, these actions occurred without verbalizations and were not sustained by Lucas. When JVM was introduced for each participant, both demonstrated a clear increase in overall play behavior (i.e., both unscripted and scripted verbalizations, and play actions).

The inclusion of peers is likely an important component of the intervention relative to unscripted responding of participants with ASD. There was variation in the frequencies of unscripted versus scripted verbalizations and play actions across typical peer partners, which may have influenced the pattern of responding of children with ASD. It is beyond the scope of this investigation to evaluate whether the verbalizations or play actions of one child in each dyad influenced the behavior of the other, but we hypothesize that each response within the script occasions the subsequent response from the partner. Should one member of the dyad alter the sequence, it is likely to lead to further deviations (i.e., unscripted responses) for the remainder of the play session. This has important implications for implementing PMI as differences in dyadic pairings may result in varied play behavior, which could be a positive outcome for many children with ASD. In addition, types of verbalizations emitted by typical peer partners (e.g., comments, questions, bids for attention, and requests) may result in differing verbalizations and play actions among children with ASD.

The acquisition of scripted responses in the present investigation deviated from previous video modeling research (D'Ateno et al. 2003; MacDonald et al. 2009; Reagon et al. 2006). All participants demonstrated clear gains in scripted

responding when the intervention was applied. However, only one of the three participants in the present investigation met mastery criteria (i.e., 75% accuracy of modeled scripts and actions). This outcome is somewhat surprising given the vast literature that shows children with ASD reliably imitate video models (Bellini and Akullian 2007). Interestingly, the scripted play outcomes are similar to those observed by Maione and Mirenda (2006), who also used multiple exemplar video models. One possible explanation for participants failing to meet mastery criteria for scripted responding is that they had a limited amount of time to engage in play sessions, and time spent engaging in unscripted verbalizations and actions meant less time to imitate the modeled script. Though mastery criteria is a common method of evaluating skill acquisition and may promote skill maintenance in behavioral interventions (see Fueller and Fienup 2018), adherence to scripts may not be the most useful method of evaluating improvements on pretend play behavior. Especially when novel, varied, and flexible responding may be considered optimal.

Saul, who showed the highest levels of unscripted play, also showed the lowest levels of scripted play. Rita showed moderate levels of scripted and unscripted play. And Lucas showed the lowest levels of unscripted play and highest levels of scripted play. Although the data suggest participants all responded positively to the JVM intervention, they responded differently, and we cannot determine why this might have been or whether this impacted the quality of play for participants with ASD.

One other potential explanation for the low levels of scripted play may have to do with the effects that multiple exemplars and intermittent exposure to videos may have had on responding. Displaying multiple videos meant participants in the current study had less total exposure to individual models compared to studies using a single video model (e.g., MacDonald et al. 2009). The rotation of three videos may have led to insufficient presentation of videos for Rita and Saul to master scripted responding. However, spontaneous pretend play in natural contexts is an area of concern for children with ASD (Kasari et al. 2006) and focusing on unscripted responding may be a more important step for interventionists. It will be important for future research to evaluate overall quality of scripted and unscripted play behaviors when modeling is used as an intervention to better understand whether the outcomes in the present investigation represent an improvement in teaching play.

Though it is unknown why there is variation in responding to video models across participants in the current study, this is commonly reported among video modeling researchers (Plavnick et al. 2015; Sherer et al. 2001). Additional features of video modeling interventions have been hypothesized to affect the likelihood that a learner will imitate a video model, such as reinforcement delivered to the model

or the target child, number of times the video model was shown, and requiring prerequisite skills for video modeling instruction (see Rayner 2015).

The present investigation also extends previous JVM studies by demonstrating implementation in the natural setting and within the early childhood classroom routine. This is the first study we know of to evaluate JVM in an early childhood setting. The outcomes therefore speak to both efficacy and potential feasibility of JVM as participants demonstrated gains in unscripted verbalizations directed toward peers in their classroom and using materials typically seen in an early childhood setting. An important step towards adoption of effective interventions by early childhood providers is feasibility within that setting (Plavnick and Ingersoll 2017). JVM was implemented successfully despite extraneous variables that could have interfered with the procedure, such as distracting noises and other typically developing peers who were interested in observing the interaction. The outcomes suggest JVM might be a procedure that could be generalized more broadly to inclusive early childhood settings.

Despite promising outcomes with regards to spontaneous pretend play observed in the present investigation, there are several limitations that could be addressed in future research. First, Rita's level of unscripted play actions during baseline makes it difficult to draw conclusions about a relation between JVM and emergence of unscripted play actions. We decided to transition her into intervention first because she demonstrated no verbalizations, and qualitatively, was observed not to respond to the typical peer's social initiations during baseline. It is difficult to say whether these were imitations of her typical peer partner or prior history of pretend play actions with similar toys. Future research might compose definitions of unscripted play that requires social interaction with a peer (e.g., turn taking) in order to better isolate social aspects of play and evaluate whether such a variable is influenced by JVM.

Second, as discussed above, Rita and Saul did not reach mastery criteria on scripted play. Although we cannot determine why this did not occur, our measurement system conservatively counted scripted responses only if performed in sequence, which may have scored less scripted behavior than actually occurred. Future research could examine measures of scripted play behavior that allow for deviation in sequence from the script. It is possible that such play is functional and capturing deviations in sequence may be more representative of skills learned.

Third, we did not collect maintenance data following the last session of intervention and cannot comment on the long-term effects of the intervention. Play behavior could have decreased following termination of the JVM intervention. Such information is important as it could guide practitioners on how long to continue using JVM. Alternatively, it is also possible that interactions could increase following

the termination of JVM. This could be due to play motivating ongoing and sustained behavior change, or the emergence of generative play resulting from interactions between unscripted and scripted play behavior performed by participants with ASD and typically developing peers. Because of the multi-component nature of JVM, with several variables potentially influencing the potency of the intervention, future research examining maintenance and additional generalization would be promising.

The present investigation showed that JVM may be an effective intervention for increasing pretend play behavior among the participants with ASD in an inclusive early childhood classroom. The use of video modeling may be an effective way to replace adult facilitated interactions between children with ASD and their peers during play, where adult-delivered prompts and reinforcement may be intrusive (Odom and Strain 1986). Consistent with previous investigations, multiple exemplar JVM may be an effective intervention to encourage novel responding in children with ASD with their typically developing peers.

Author Contributions ADD conceived the study, designed, collected data and coordinated and drafted the manuscript; JBP participated in the design and interpretation of the data and edited the manuscript; MYSB participated in the design and interpretation of data and calculated the effect sizes and wrote that section of manuscript. All authors read and approved the final manuscript.

Funding This study was not funded.

Compliance with Ethical Standards

Conflict of interest The authors declare no conflict of interest.

Ethical Approval All procedures performed in studies involving human participants were in accordance with the Ethical Standards of the Institutional and/or National Research Committee and with the 1964 Helsinki Declaration and its later amendments or comparable ethical standards.

Informed Consent Informed consent was obtained from all individual participants included in the study.

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